



Biology of Blood and Marrow Transplantation

journal homepage: www.bbmt.org



Clinical Research: Supportive Care

Impact of Gut Colonization by Antibiotic-Resistant Bacteria on the Outcomes of Allogeneic Hematopoietic Stem Cell Transplantation: A Retrospective, Single-Center Study



Jaroslav Bilinski¹, Katarzyna Robak¹, Zinaida Peric², Halina Marchel³, Ewa Karakulska-Prystupiuik¹, Kazimierz Halaburda¹, Patrycja Rusicka¹, Ewa Swoboda-Kopec³, Marta Wroblewska^{3,4}, Wieslaw Wiktor-Jedrzejczak¹, Grzegorz W. Basak^{1,*}

¹ Department of Hematology, Oncology and Internal Diseases, Medical University of Warsaw, Warsaw, Poland

² Department of Hematology, Internal Clinic, University Hospital Centre Zagreb, Zagreb, Croatia

³ Department of Microbiology, Central Clinical Hospital, Medical University of Warsaw, Warsaw, Poland

⁴ Department of Dental Microbiology, Medical University of Warsaw, Warsaw, Poland

Article history:

Received 12 November 2015

Accepted 12 February 2016

Key Words:

Gut colonization
Antibiotic-resistant bacteria
Hematopoietic stem cell transplantation
Bacteremia

A B S T R A C T

Gut colonization by antibiotic-resistant bacteria may underlie hard-to-treat systemic infections. There is also accumulating evidence on the immunomodulatory function of gut microbiota after allogeneic stem cell transplantation (alloSCT) and its impact on graft-versus-host disease (GVHD). We investigated the epidemiology and clinical impact of gut colonization after alloSCT and retrospectively analyzed data on 107 alloSCTs performed at a single transplant center. Pretransplant microbiology screening identified colonization in 31% of cases. Colonization had a negative impact on overall survival after alloSCT in univariate (34% versus 74% at 24 months, $P < .001$) and multivariate (hazard ratio, 3.53; 95% confidence interval, 1.71 to 7.28; $P < .001$) analyses. Nonrelapse mortality was significantly higher in colonized than in noncolonized patients (42% versus 11% at 24 months, $P = .001$). Colonized patients more frequently experienced bacteremia (48% versus 24%, $P = .01$), and more deaths were attributable to infectious causes in the colonized group (42% versus 11% of patients and 67% versus 29% of deaths, $P < .05$). We observed a significantly higher incidence of grades II to IV acute GVHD in colonized than in noncolonized patients (42% versus 23%, $P < .05$), especially involving the gastrointestinal system (33% versus 13.5%, $P = .07$). In summary, we determined that gut colonization by antibiotic-resistant bacteria decreases the overall survival of patients undergoing alloSCT by increasing nonrelapse mortality and the incidences of systemic infection and acute GVHD.

© 2016 The American Society for Blood and Marrow Transplantation.

INTRODUCTION

The incidence of infections by antibiotic-resistant bacteria (ARB) is rising because of selection pressure caused by increasing use of broad-spectrum antibiotics [1]. The field of hematology is particularly affected, because patients with neoplastic blood disorders undergo frequent antimicrobial prophylaxis and treatment [2]. Pathogenic bacteria can colonize niches in the human body, including the gastrointestinal tract [3], whereas chemotherapy-induced damage to the gut epithelium and overlap neutropenia enable bacteria

to spread through the gut wall, causing life-threatening systemic infections [4,5]. In patients who have undergone allogeneic stem cell transplantation (alloSCT), systemic infections with ARB are associated with a mortality rate of 36% to 95% [4,6,7].

To our knowledge, no study has addressed the epidemiology of gut colonization by ARB in patients undergoing alloSCT or its impact on the outcomes of the procedure. We hypothesized that colonization would increase the incidence of bacterial infections after alloSCT, thereby increasing nonrelapse mortality (NRM) and decreasing overall survival (OS). We also postulated that gut colonization and associated systemic infection would influence the immune system after alloSCT, affecting the incidence of graft-versus-host disease (GVHD). To test these assumptions, we retrospectively analyzed the outcomes of allogeneic transplantations

Financial disclosure: See Acknowledgments on page 1093.

* Correspondence and reprint requests: Grzegorz W. Basak, MD, PhD, Department of Hematology, Oncology and Internal Diseases, Medical University of Warsaw, 1a Banacha Str., 02-097 Warsaw, Poland.

E-mail address: grzegorz.basak@wum.edu.pl (G.W. Basak).

performed in 107 patients at a single transplant center according to gut colonization status.

METHODS

Study Design

This retrospective, single-center study was performed using clinical data on consecutive patients who underwent alloSCT between 2010 and 2013 in the Department of Hematology, Oncology and Internal Diseases of the Medical University of Warsaw, Poland. All patients were routinely screened for gut colonization by the following ARB: methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin-resistant *Enterococcus* (VRE), extended-spectrum β -lactamase (ESBL)-producing Enterobacteriaceae, and carbapenemase-producing Enterobacteriaceae.

Colonization status was ascertained by the microbiologic examination of rectal swabs cultured on the following chromogenic media: MRSA Agar (Graso, Gdańsk, Poland), VRE Agar (Graso), chromID ESBL (bioMérieux S.A., Marcy l'Etoile, France), and CHROMagar *Klebsiella pneumoniae* carbapenemase (Graso). Pathogens were streaked on Columbia Agar plates with 5% sheep blood (Becton, Dickinson and Company, Franklin Lakes, NJ) and identified using a MALDI Biotyper (Bruker Daltonics Inc., Billerica, MA). MRSA, VRE, ESBL, and carbapenemase-producing Enterobacteriaceae were verified using a phenotypic method, according to national recommendations [8–10]. Additionally, the ability of isolates to produce carbapenemases (metallo- β -lactamase, *K. pneumoniae* carbapenemase, expanded-spectrum oxacillinase-48) was investigated using the Rapidec Carba NP biochemical assay (bioMérieux S.A.) and/or the GeneXpert qualitative real-time PCR method (Cepheid, Sunnyvale, CA). The pathogens identified were designated ARB [11]. Colonization was defined as the detection of an antibiotic-resistant microorganism in at least 1 rectal swab. Patients colonized with at least 1 ARB before alloSCT were termed “colonized,” whereas the remaining patients were termed “noncolonized.”

Transplants were conducted according to local procedures and treatment protocols. Blood cultures were performed during every febrile episode and repeated depending on clinical indications. Microbiologic blood testing was undertaken using standard techniques. For patients on antibiotics, Plus Aerobic/F and Plus Anaerobic/F culture bottles were used (Becton, Dickinson and Company) and monitored using the Bactec FX system (Becton, Dickinson and Company) for 7 days. Microorganisms were identified using a MALDI Biotyper (Bruker Daltonics Inc.). Antimicrobial susceptibility was tested and interpreted according to current European Committee on Antimicrobial Susceptibility Testing recommendations [12].

Bacteremia was defined as at least 1 positive blood culture, irrespective of clinical symptoms. To diagnose bacteremia caused by coagulase-negative pathogens, 2 consecutive positive cultures were required. In case of febrile episodes during the neutropenic period, the transplantation protocol included the use of antibiotics targeting the gut-colonizing ARB in addition to standard, broad-spectrum antibiotics. Toxicities, including mucositis, were graded and modified using the toxicity criteria of the Common Terminology Criteria for Adverse Events, version 4.0 [13]. GVHD was diagnosed and graded according to standard criteria [14,15]. Transplant risk was assessed using the updated European Society for Blood and Marrow Transplantation (EBMT) risk score [16]. Low-risk disease before alloSCT was defined as early-stage disease, and high-risk disease was defined as intermediate- and late-stage disease, according to EBMT definitions.

All patients provided standard informed consent for alloSCT, data analysis, and publication. Collection of rectal swabs was performed as a standard practice in all hospitalized patients according to the local guidelines. According to Polish law, the approval of the institutional review board was not required because of the noninterventional nature of the study.

Statistical Analysis

The primary endpoint of the study was OS after alloSCT, which was defined as the probability of survival after transplantation, irrespective of disease state, at any point in time. Surviving patients were censored at the last follow-up examination or at the date of subsequent alloSCT. NRM was defined as the time to death without relapse/progression. Deaths from any cause without prior progression were considered events, whereas relapse or progression was considered a competing event. Patients were censored at last follow-up or, in case of relapse/progression, on the date of relapse/progression. Relapse incidence (RI) was defined as the time to relapse or progression. Deaths without relapse or progression were considered competing events.

Patients' baseline characteristics were summarized and tested with appropriate statistical analyses (the *t*-test and Mann-Whitney U-test for continuous variables and the chi-square test and Fisher's exact test for categorical variables). NRM and RI were calculated using the cumulative incidence method. Survival estimates were calculated using the Kaplan-Meier method, and curves were compared using the log-rank test. The

association between time to death and colonization status and other relevant variables was evaluated in a multivariate analysis using Cox's proportional hazards regression analysis. All statistical analyses were performed using SPSS versions 18.0 and 19.0 (SPSS Inc., Chicago, IL, and IBM Corp., Armonk, NY, respectively) and R version 2.12.2 with the package “survival” (<http://www.r-project.org/foundation>).

RESULTS

Patient Characteristics

The study included 107 patients who underwent 115 alloSCTs. Eight cases of second and third transplantations were excluded from the analysis. Clinical characteristics are shown in Table 1. Briefly, most patients were male (51%) with a median age of 47 years (range, 18 to 78). Acute leukemia was the most common diagnosis (53%). The median time from diagnosis to transplantation was 10.8 months (range, 2 to 241). Conditioning was either myeloablative (61%) or reduced intensity (39%). Thirty-eight percent of transplants were performed from matched related donors, 47% from matched unrelated donors, and 16% from mismatched unrelated donors. Peripheral blood was the source of the transplant in most cases (94%), and transplantation from female donors to male recipients occurred in 31% of cases. The average EBMT transplant risk score was 4 (range, 1 to 7). GVHD prophylaxis included antithymocyte globulin (50%), cyclosporine A (97%), methotrexate (80%), or mycophenolate mofetil (20%). There were no differences in transplant characteristics between the colonized and noncolonized groups (Table 1).

At the time of alloSCT, 31% of patients were colonized by ARB: 21% by vancomycin-resistant Gram-positive bacteria, 20% by ESBL-producing Gram-negative rods, and 6% by strains of Gram-negative bacteria resistant to carbapenems (Figure 1). Eighteen patients were colonized by more than 1 (up to 4) antibiotic-resistant pathogen. Moreover, 14% of patients were identified as past carriers of ARB (up to 1 year before but not at the time of alloSCT), and their colonization characteristics and outcomes are shown in Supplementary Table 1.

Outcomes

After transplantation, neutrophil engraftment was observed after a median of 18 days (range, 11 to 101) and platelet engraftment after a median of 13 days (range, 0 to 102; Table 2). Grades 3 to 4 mucositis occurred in 24% of patients, and its incidence did not differ between colonized and noncolonized patients. Acute GVHD (aGVHD) was observed in 48% of patients; grades II to IV aGVHD occurred in 29% of patients and was more frequent in colonized (42%) than in noncolonized (23%) patients ($P < .05$). Organ manifestations involved the gastrointestinal tract in 40%, liver in 4%, and skin in 61% of patients with aGVHD. Colonized patients developed gastrointestinal manifestations of aGVHD more often than noncolonized patients (33% versus 13.5%, $P = .07$). Chronic GVHD developed in 52% and extensive chronic GVHD in 18% of patients, and there was no difference in their incidences between the groups (Table 2).

In the early post-transplant period, 69% of patients experienced infections, which were confirmed to be bacterial infections in 53%. Overall, 74% of patients required blood sampling for microbiologic testing; the sampling frequency was similar between colonized and noncolonized patients. A total of 294 blood cultures was performed, with a median of 2 blood cultures per patient (range, 1 to 18). In 32% of patients at least 1 blood culture was positive, whereas in 16% of the total ARB were detected. The most common causes of bacteremia were coagulase-negative *Staphylococcus* spp. (37% of positive

Table 1
Baseline Demographic Characteristics of the Study Population

Characteristics	All Patients (n = 107)	Noncolonized (n = 74)	Colonized (n = 33)	P
Year of alloSCT, median (range)	2012 (2010–2013)	2012 (2010–2013)	2012 (2010–2013)	NS
Male sex	55 (51)	39 (53)	16 (48)	NS
Age at alloSCT, yr, median (range)	47 (18–78)	47 (18–65)	48 (20–78)	NS
Time from diagnosis to alloSCT, mo, median (range)	10.8 (2–241)	10.6 (2–206)	10.8 (2.5–241)	NS
Diagnosis				
AML	48 (45)	35 (47)	143 (39)	NS
ALL	9 (8)	4 (5)	5 (15)	
MPD	8 (7)	6 (8)	2 (6)	
MDS	11 (10)	7 (9)	4 (12)	
Lymphoma	14 (13)	10 (14)	4 (12)	
MDS/MPD	4 (4)	4 (5)	0	
Other	13 (12)	8 (11)	5 (15)	
Disease risk at alloSCT*				
Low	39 (58)	30 (40)	9 (27)	NS
High	52 (27)	34 (46)	18 (55)	
Unknown	16 (15)	10 (14)	6 (18)	
EBMT risk score, median (range)	4 (1–7)	4 (1–7)	4 (1–6)	NS
Conditioning: myeloablative/RIC	65/42 (61/39)	46/28 (62/38)	19/14 (58/42)	NS
Stem cell source: PBSCT/BMT	100/2 (94/2)	67/2 (92/3)	33/0 (100)	NS
WBC before conditioning, $\times 10^9/L$	3.9 (.1–119.5)	3.95 (.1–79)	3.3 (.3–119.5)	NS
ANC before conditioning, $\times 10^9/L$	2.1 (0–74)	2.2 (0–74)	1.35 (0–31)	.06
Stem cell donor				
MRD	39 (38)	28 (40)	11 (33)	NS
MUD	48 (47)	32 (46)	16 (48)	
MMUD	16 (16)	10 (14)	6 (18)	
Sex mismatch, female to male	33 (31)	27 (37)	6 (18)	.08
MTX post-alloSCT	84 (80)	59 (80)	25 (7)	NS

NS indicates nonsignificant ($P \geq .1$); AML, acute myeloblastic leukemia; ALL, acute lymphoblastic leukemia; MPD, myeloproliferative disorders; MDS, myelodysplastic syndrome; RIC, reduced-intensity conditioning; PBSCT, peripheral blood stem cell transplantation; BMT, bone marrow transplantation; ANC, absolute neutrophil count; MRD, matched related donor; MUD, matched unrelated donor; MMUD, mismatched unrelated donor; MTX, methotrexate.

Values are number of cases with percents in parentheses, unless otherwise noted.

* Low-risk group means early disease stage and high-risk means intermediate and late disease stage according to EBMT transplant risk score (see [Methods](#)).

cultures), followed by VRE (17%) and *K. pneumoniae* ESBL⁺ (15%; [Figure 1](#)). A positive blood culture was observed more often in colonized than in noncolonized patients (48% versus 24%, $P = .01$), and ARB bacteremia occurred significantly more

frequently in colonized than in noncolonized patients (30% versus 9% of patients, $P < .01$). Notably, only 5 colonized patients (14%) experienced confirmed systemic infection with gut-colonizing microorganisms: 3 of 9 patients (33%) who

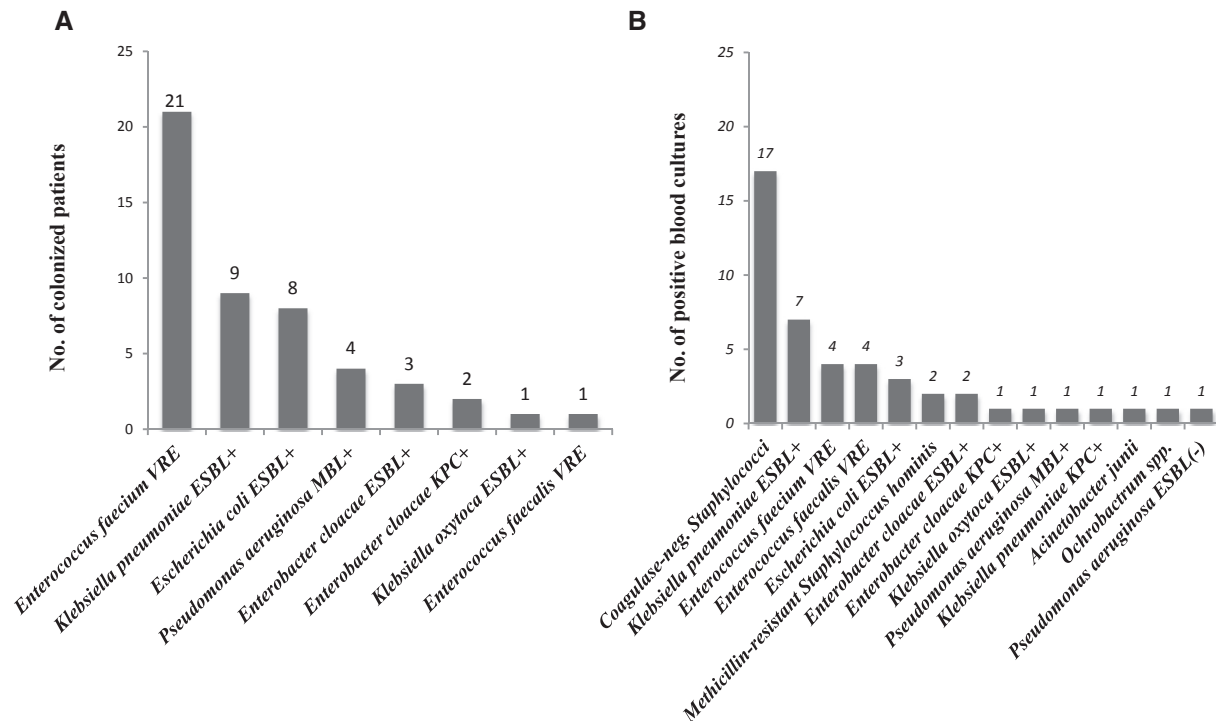


Figure 1. Frequencies of bacterial species isolated from (A) rectal swabs collected at the time of alloSCT (colonization) and (B) blood taken after alloSCT (bacteremia). MBL indicates metallo- β -lactamase; KPC, *Klebsiella pneumoniae* carbapenemase.

Table 2
Transplant-Related Events and Outcomes

Variable	All Patients (n = 107)	Noncolonized (n = 74)	Colonized (n = 33)	P
ANC > .5 × 10 ⁹ /L, days, median (range)	18 (11–101)	18 (11–101)	19 (11–101)	NS
PLT > 20 × 10 ⁹ /L, days, median (range)	13 (0–102)	13 (0–102)	13 (0–101)	NS
Mucositis	64 (60)	43 (58)	21 (64)	NS
Grades III–IV	26 (24)	18 (24)	8 (24)	NS
aGVHD	51 (48)	31 (42)	20 (61)	NS
aGVHD grades II–IV	31 (29)	17 (23)	14 (42)	<.05
GI aGVHD	21 (20)	10 (13.5)	11 (33)	.07
cGVHD	57 (52)	41 (51)	17 (49)	NS
Extensive	19 (18)	14 (19)	5 (15)	NS
Relapse	24 (22)	17 (23)	7 (21)	NS
Blood cultures				
Patients with ≥1 performed	79 (74)	51 (69)	28 (85)	NS
Number per patient, average (range)	2.8 (0–18)	2.5 (0–18)	3.3 (0–18)	NS
Patients with positive blood culture	34 (32)	18 (24)	16 (48)	.01
Patients with ARB in blood culture	17 (16)	7 (9)	10 (30)	<.01
Cause of death (% of all patients; % of all deaths)				
Infection	22 (21; 45)	8 (11; 29)	14 (42; 67)	<.05
Relapse/progression	17 (16; 35)	14 (19; 50)	3 (9; 14)	
GVHD	6 (6; 12)	3 (4; 11)	3 (9; 14)	
Other*	3 (3; 6)	2 (3; 7)	1 (3; 5)	
Unknown	1 (1; 2)	1 (1; 4)	0	
OS, % (95% CI)				
At 3 mo	87 (81–94)	94 (88–99)	77 (63–94)	<.001
At 12 mo	72 (64–82)	81 (72–91)	53 (38–75)	
At 24 mo	62 (52–73)	74 (64–86)	34 (19–60)	
NRM, % (95% CI)				
At 3 mo	9 (5–16)	6 (2–14)	16 (6–32)	.001
At 12 mo	18 (11–26)	9 (4–18)	37 (20–54)	
At 24 mo	20 (13–29)	11 (5–20)	42 (23–60)	

PLT indicates platelets; GI, gastrointestinal; cGVHD, chronic graft-versus-host disease.

Values are number of cases with percents in parentheses, unless otherwise noted.

* Graft failure, heart failure, toxicity of conditioning.

were colonized with *K. pneumoniae* ESBL⁺ and 2 of 21 patients (9%) who were colonized with *Enterococcus faecium* VRE. *K. pneumoniae* ESBL⁺ bacteremia occurred significantly more frequently in patients colonized with this bacterium (33%) compared with noncolonized patients (4%, $P < .001$), whereas there was no significant difference in the incidence of *E. faecium* VRE bacteremia between the groups. Similar analyses of transplant outcomes were performed in the group of patients not colonized at the time of alloSCT but colonized within 1 year before alloSCT ($n = 15$, past colonizers). The results are shown in [Supplementary Table 1](#). Although the statistical comparisons with remaining patients could not be done because of a low number of observations, it seems that the outcomes of transplantation in this group are comparable with noncolonized patients. Notably, 2 past colonizers developed bacteremia with gut-colonizing organism previously identified (*K. pneumoniae* ESBL⁺ and *E. faecium* VRE).

During their inpatient stay, piperacillin-tazobactam was used in 56%, meropenem in 58%, linezolid in 27%, and colistin in 6% of patients. When analyzed separately, a significantly higher proportion of colonized patients received linezolid (42% versus 20% of noncolonized, $P = .05$),

whereas colistin was used exclusively in patients colonized with carbapenem-resistant strains. Meropenem and piperacillin-tazobactam were used in comparable proportions of colonized and noncolonized patients (70% versus 55% and 55% versus 59%, both nonsignificant, respectively).

Kaplan-Meier estimates of OS for all patients and for each of the groups are shown in [Figure 2](#). Forty-nine patients (46%) died. The median length of follow-up was 26.5 months (range, 10 to 56). The estimated rate of OS for all patients was 87% (95% confidence interval [CI], 81% to 94%) at 3 months, 72% (95% CI, 64% to 82%) at 12 months, and 62% (95% CI, 52% to 73%) at 24 months after transplantation. It was significantly lower in the colonized group than in the noncolonized group (77% versus 94% at 3 months, 53% versus 81% at 12 months, and 34% versus 74% at 24 months, $P < .001$; [Figure 2B](#), [Table 2](#)). The standard risk factors for alloSCT (age, donor type, intensity of conditioning, stage of disease) were associated with a less pronounced impact on OS than bacterial colonization status ([Figure 2](#)). To determine the impact of colonization on OS after alloSCT corrected for other potentially confounding factors, we performed a multivariate analysis ([Table 3](#)): Only gut colonization had a significant impact on OS (hazard ratio, 3.53; 95% CI, 1.71 to 7.28; $P < .001$).

The estimated incidence of NRM in all patients was 9% (95% CI, 5% to 16%) after 3 months, 18% (95% CI, 11% to 26%) after 12 months, and 20% (95% CI, 13% to 29%) after 24 months ([Figure 3](#), [Table 2](#)). Colonized patients exhibited a significantly higher NRM than noncolonized patients: 16% (95% CI, 6% to 32%) versus 6% (95% CI, 2% to 14%) after 3 months, 37% (95% CI, 20% to 54%) versus 9% (95% CI, 4% to 18%) after 12 months, and 42% (95% CI, 23% to 60%) versus 11% (95% CI, 5% to 20%) after 24 months ($P = .001$).

The RI was 5% (95% CI, 2% to 11%) after 3 months, 14% (95% CI, 8% to 22%) after 12 months, and 21% (95% CI, 14% to 31%) after 24 months; there were no differences in RI between the groups ([Figure 3](#)).

Regarding causes of death, infection dominated (45%), followed by relapse/progression (35%) and GVHD (12%). The distribution of causes of death differed between colonized and noncolonized groups. Infection was responsible for 67% versus 29%, relapse for 14% versus 50%, and GVHD for 14% versus 11% of deaths in colonized and noncolonized patients, respectively ($P < .05$). Only 4 patients (4%) died of bacterial sepsis caused by ARB (*K. pneumoniae* ESBL⁺, *K. pneumoniae* carbapenemase-positive, *E. faecium* VRE, and *Pseudomonas aeruginosa* metallo- β -lactamase-positive). All 4 were colonized by these bacteria within 1 year before alloSCT, although 2 had negative colonization status at transplantation (*K. pneumoniae* ESBL⁺ and *E. faecium* VRE).

DISCUSSION

In this retrospective, single-center analysis, we identified gut colonization with at least 1 ARB in about one third of patients undergoing alloSCT. Most frequently, patients were colonized by VRE or Enterobacteriaceae ESBL⁺; however, a proportion was colonized by as many as 4 different ARB. Gut colonization in these patients probably resulted from the development of drug-resistance mechanisms under the strong pressure exerted by numerous lines of antibiotic treatment or the acquisition of new gut-colonizing pathogens via contact with other patients and medical staff.

Our study revealed that estimated OS after alloSCT is over 2-fold lower in colonized patients than in noncolonized patients (34% versus 74% at 2 years). Gut colonization was the only factor that significantly affected

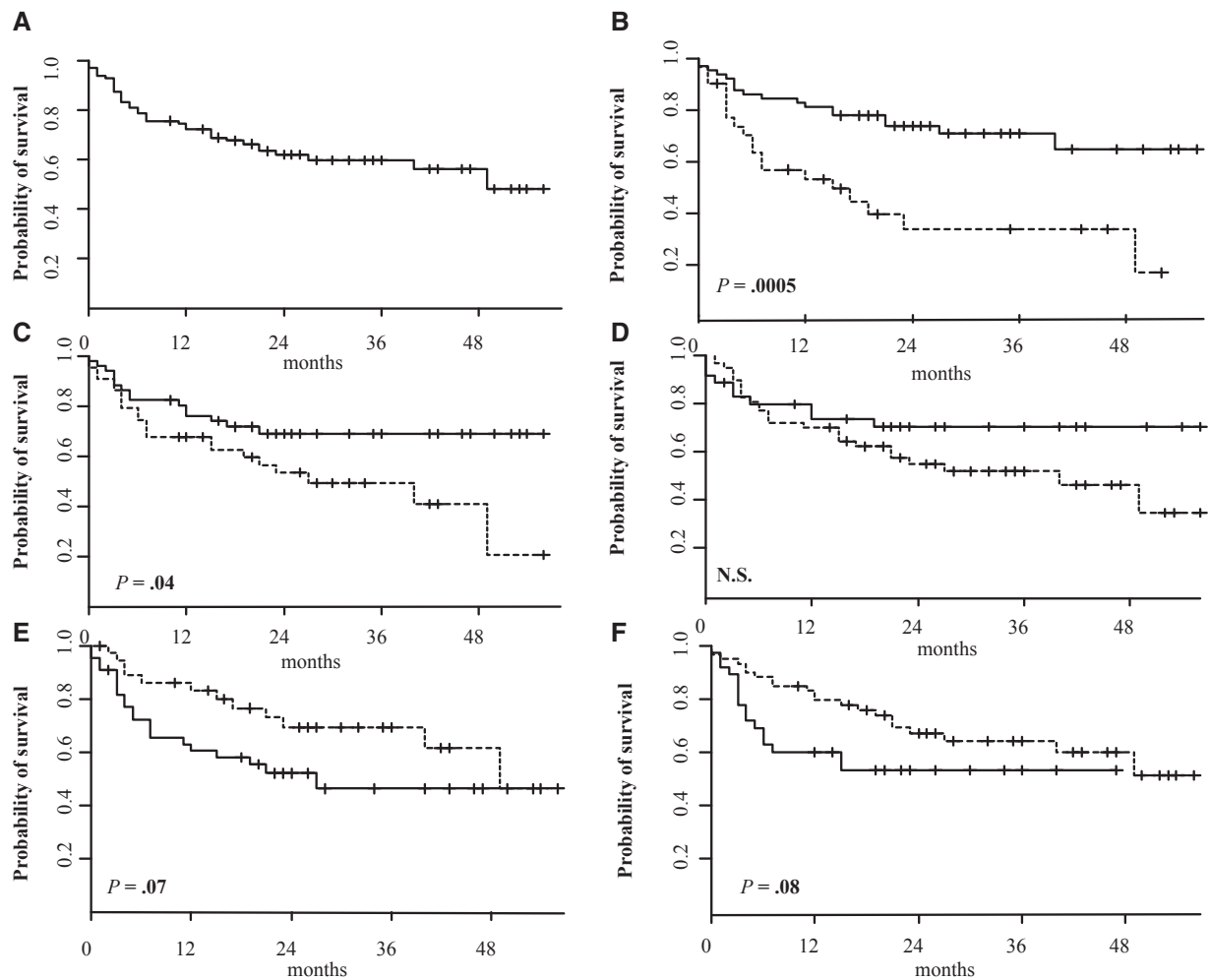


Figure 2. Overall patient survival after alloSCT. (A) All patients. (B) Stratified by colonized (dashed line) versus noncolonized (solid line): 34% (95% CI, 19% to 60%) versus 74% (95% CI, 64% to 86%), $P < .001$. (C) Stratified by age > 50 (dashed line) versus <50 years (solid line): 69% (95% CI, 57% to 84%) versus 21% (95% CI, 4% to 89%), $P < .05$. (D) Stratified by unrelated donor (dashed line) versus related donor (solid line): 70% (95% CI, 56% to 88%) versus 34% (95% CI, 18% to 67%), nonsignificant. (E) Stratified by low-risk disease (dashed line) versus high-risk disease (solid line): 46% (95% CI, 32% to 67%) versus 46% (95% CI, 24% to 89%), $P = .07$. (F) Stratified by myeloablative conditioning (dashed line) versus reduced-intensity conditioning (solid line): 53% (95% CI, 39% to 73%) versus 51% (95% CI, 35% to 76%), $P = .08$, at 24 months.

survival. Moreover, colonization showed a strong effect on OS after multivariate adjustment for other clinical predictors, with a hazard ratio of 3.53 for positive colonization status, whereas the presence of high-risk disease at alloSCT was associated with a lower risk of mortality. Although the negative impact of gut colonization by pathogenic bacteria on OS after alloSCT seems obvious, to our knowledge this has not been documented to date. Our observations are consistent with those of Taur et al. [17], who reported that loss of heterogeneity of the gut microbiome contributes to a significantly decreased OS after alloSCT. Given that gut colonization is only enabled in patients with a poor microbiome repertoire [18–20], it is possible that gut colonization

serves as a surrogate marker of a poor gut microbiome. However, this assumption requires further study, including complex analysis of bacterial microbiome.

In this study there were no differences in the characteristics of patients in the 2 groups, and RI was also comparable. Conversely, NRM was significantly higher in colonized patients than in noncolonized patients (42% versus 11% at 2 years). We hypothesize that this was attributable to a higher incidence of hard-to-treat infections. Infections were the most common cause of death, and the proportion of deaths caused by infection was significantly higher in the colonized than in the noncolonized group (67% versus 29% of all deaths). This higher incidence of systemic bacterial infections was also reflected in the significantly higher proportion of positive blood cultures (48% versus 24%) and cultures with an identified ARB (30% versus 9%, $P < .01$) in the colonized group than in the noncolonized group. Interestingly, there was almost no overlap between the colonizing bacteria and pathogen identified by blood culture. Overlap was found only for *K. pneumoniae* ESBL⁺, for which positive blood cultures were identified in 33% of colonized patients and 4% of noncolonized patients ($P < .001$). This contrasts

Table 3
Multivariate Analysis of Predictors of OS after AlloSCT

Risk Factor	Hazard Ratio	95% CI	P
Colonized patient	3.53	1.71–7.28	.0006
Age of recipient > 50 yr	1.21	.58–2.52	NS
High-risk disease	2.00	.96–4.22	.07
Unrelated donor	1.56	.71–3.44	NS
Myeloablative conditioning	.59	.28–1.26	NS

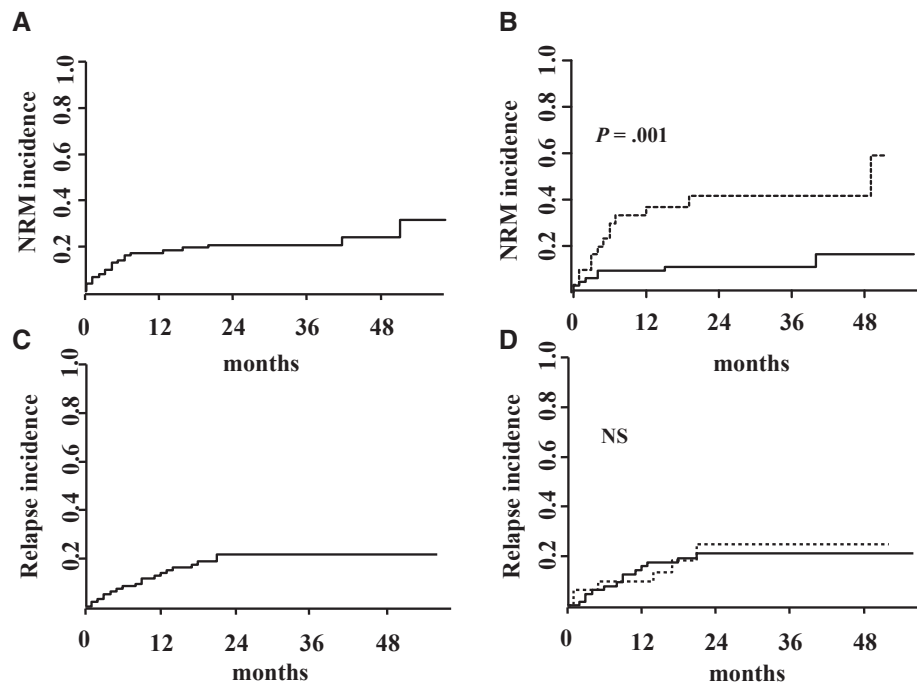


Figure 3. NRM after alloSCT in (A) all patients: 20% (95% CI, 13% to 29%) at 24 months, and (B) stratified by colonized (dashed line) versus noncolonized (solid line): 42% (95% CI, 23% to 60%) versus 11% (95% CI, 5% to 20%; $P = .001$) at 24 months. RI after alloSCT in (C) all patients: 22% (95% CI, 14% to 31%) after 24 months, and (D) stratified by colonized (dashed line) versus noncolonized (solid line): 21% (95% CI, 12% to 32%) versus 24% (95% CI, 9% to 43%; nonsignificant) at 24 months.

with data from published studies in which the frequency of systemic infections caused by pathogens previously colonizing nonsterile body sites is considerable and dependent on the pathogen: 58% for *P. aeruginosa* multidrug-resistant (MDR) [21], 40% for carbapenem-resistant *K. pneumoniae*, 47% for Enterobacteriaceae ESBL+ [22], and 2% for VRE [23]. The differences may depend on the low sensitivity of the traditional blood culture techniques used but are more likely to reflect different standards of empiric antibiotic treatment in patients with confirmed colonization. At our center colonized patients routinely receive antibiotics targeted at gut-colonizing bacteria during every febrile episode when neutropenia is present; therefore, the results of repeated blood cultures are likely to be negative.

There is growing evidence that the gut microbiome exerts immunomodulatory effects during alloSCT [24,25]. Traditionally, it was thought that anaerobic bacteria in the gut contribute to the development of GVHD, because decontamination with metronidazole before alloSCT was found to contribute to a decreased incidence and severity of GVHD [26]. Given that colonization is usually related to an impaired gut microbiome, we hypothesized that it could affect the development of GVHD. Our analyses revealed that colonized patients more frequently developed clinically relevant (grades II to IV) aGVHD (42% versus 23%, $P < .05$) and that about twice as many colonized patients developed gut manifestations of aGVHD compared with noncolonized patients (33% versus 13.5%, $P = .07$). This effect was not observed for chronic GVHD. It is possible to argue that colonized patients suffer more frequently from *Clostridium difficile* colitis, which has similar symptoms to intestinal GVHD. However, a stool analysis for *C. difficile* toxin and stool culture are routinely performed in all patients with post-transplant gastrointestinal symptoms at our center, and in cases described

here, the diagnosis of isolated intestinal GVHD was made only in patients negative for *C. difficile* infection. It is more plausible that the more frequent infections in colonized patients could have an adjuvant effect and trigger aGVHD via the activation of alloreactive lymphocytes. Furthermore, if gut colonization reflects a poor and pathologic gut microbiome that does not fully support the immunoregulatory functions of commensal bacteria—such as the development of regulatory T cells, balancing of nutrients, and provision of short-chain fatty acids—alloimmune responses may be triggered [24,27].

Our data suggest that gut colonization with ARB is associated with a significantly decreased OS because of an increased NRM. The factors responsible for this include an increased rate of bacterial infections and a higher incidence of aGVHD. It should be noted, however, that despite numerous studies linking the intestinal microbiome with the OS, NRM or GVHD in alloSCT patients it has not been proven a direct cause-and-effect link, and these associations should be regarded as hypotheses. Patients who were colonized could be potentially those who had serious infections during previous chemotherapy and spent more time in the hospital, enabling colonization with resistant flora. Our study may stimulate discussion on the role of gut colonization as a factor to be considered during the selection of patients for alloSCT. In our opinion, it is worthwhile to routinely screen patients for gut colonization with ARB before alloSCT. Unfortunately, even empiric antibiotic therapy targeted at colonizing microorganisms during febrile episodes post-alloSCT did not protect the group of colonized patients from the detrimental effects of colonization on survival after transplantation. Therefore, our observations support the need for decolonization strategies before alloSCT involving, for instance, the restoration of healthy gut flora by the transplantation of fecal microbiota.

ACKNOWLEDGMENTS

The authors thank all the physicians, nurses, and laboratory staff for their invaluable contribution to the current project, especially Dr. Emilian Snarski for design of the clinical data capture system.

Financial disclosure: This work was supported by statutory funds of the Department of Hematology, Oncology and Internal Diseases, the Medical University of Warsaw, Warsaw, Poland.

Conflict of interest statement: There are no conflicts of interest to report.

SUPPLEMENTARY DATA

Supplementary data related to this article can be found online at <http://dx.doi.org/10.1016/j.bbmt.2016.02.009>.

REFERENCES

- Roca I, Akova M, Baquero F, et al. The global threat of antimicrobial resistance: science for intervention. *New Microbes New Infect.* 2015; 6:22–29.
- Mikulska M, Viscoli C, Orasch C, et al. Aetiology and resistance in bacteraemias among adult and paediatric haematology and cancer patients. *J Infect.* 2014;68:321–331.
- Walker A. Intestinal colonization and programming of the intestinal immune response. *J Clin Gastroenterol.* 2014;48(Suppl 1):S8–S11.
- Girmenia C, Rossolini GM, Picocchi A, et al. Infections by carbapenem-resistant *Klebsiella pneumoniae* in SCT recipients: a nationwide retrospective survey from Italy. *Bone Marrow Transplant.* 2015;50: 282–288.
- Oostdijk EA, Kesecioglu J, Schultz MJ, et al. Effects of decontamination of the oropharynx and intestinal tract on antibiotic resistance in ICUs: a randomized clinical trial. *JAMA.* 2014;312:1429–1437.
- Kim SB, Min YH, Cheong JW, et al. Incidence and risk factors for carbapenem- and multidrug-resistant *Acinetobacter baumannii* bacteremia in hematopoietic stem cell transplantation recipients. *Scand J Infect Dis.* 2014;46:81–88.
- Caselli D, Cesaro S, Ziino O, et al. Multidrug resistant *Pseudomonas aeruginosa* infection in children undergoing chemotherapy and hematopoietic stem cell transplantation. *Haematologica.* 2010;95: 1612–1615.
- Zabicka D, Hryniewicz W. Recommendations on the selection of tests to determine the susceptibility of bacteria to antibiotics and chemotherapeutics 2010 [Rekomendacje doboru testów do oznaczania wrażliwości bakterii na antybiotyki i chemioterapeutyki 2010]. Retrieved from www.korl.edu.pl. Accessed September 22, 2015.
- Gniadkowski M, Zabicka D, Hryniewicz W. Recommendations on the selection of tests to determine the susceptibility of bacteria to antibiotics and chemotherapeutics 2009. Determination of sensitivity of Gram-negative bacteria [Rekomendacje doboru testów do oznaczania wrażliwości bakterii na antybiotyki i chemioterapeutyki 2009. Oznaczanie wrażliwości pałeczek Gram-ujemnych]. Retrieved from www.korl.edu.pl. Accessed September 22, 2015.
- Zabicka D, Baraniak A, Gniadkowski M, Hryniewicz W. Detection of carbapenemases - recommendations 2013 [Wykrywanie karbapenemaz - zalecenia 2013]. Retrieved from www.korl.edu.pl. Accessed September 22, 2015.
- Magiorakos AP, Srinivasan A, Carey RB, et al. Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. *Clin Microbiol Infect.* 2012;18:268–281.
- European Committee on Antimicrobial Susceptibility Testing (EUCAST). Breakpoint tables for interpretation of MICs and zone diameters. Version 5.0 ed, 2015. Retrieved from www.eucast.org/clinical_breakpoints/. Accessed September 24, 2015.
- U.S. Department of Health and Human Services, National Institutes of Health, National Cancer Institute. Common terminology criteria for adverse events (CTCAE). v4.03 2010. Retrieved from www.evs.nci.nih.gov. Accessed September 24, 2015.
- Glucksberg H, Storb R, Fefer A, et al. Clinical manifestations of graft-versus-host disease in human recipients of marrow from HL-A-matched sibling donors. *Transplantation.* 1974;18:295–304.
- Shulman HM, Sullivan KM, Weiden PL, et al. Chronic graft-versus-host syndrome in man. A long-term clinicopathologic study of 20 Seattle patients. *Am J Med.* 1980;69:204–217.
- Gratwohl A. The EBMT risk score. *Bone Marrow Transplant.* 2012;47: 749–756.
- Taur Y, Jenq RR, Perales MA, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood.* 2014;124:1174–1182.
- Ubeda C, Bucci V, Caballero S, et al. Intestinal microbiota containing *Barnesiella* species cures vancomycin-resistant *Enterococcus faecium* colonization. *Infect Immun.* 2013;81:965–973.
- Freter R. The fatal enteric cholera infection in the guinea pig, achieved by inhibition of normal enteric flora. *J Infect Dis.* 1955;97:57–65.
- Bohnhoff M, Miller CP, Martin WR. Resistance of the mouse's intestinal tract to experimental salmonella infection. II. Factors responsible for its loss following streptomycin treatment. *J Exp Med.* 1964;120:817–828.
- Liss BJ, Vehreschild JJ, Cornely OA, et al. Intestinal colonisation and blood stream infections due to vancomycin-resistant enterococci (VRE) and extended-spectrum beta-lactamase-producing Enterobacteriaceae (ESBLE) in patients with haematological and oncological malignancies. *Infection.* 2012;40:613–619.
- Tumbarello M, Spanu T, Sanguinetti M, et al. Bloodstream infections caused by extended-spectrum-beta-lactamase-producing *Klebsiella pneumoniae*: risk factors, molecular epidemiology, and clinical outcome. *Antimicrob Agents Chemother.* 2006;50:498–504.
- Kang CI, Song JH, Chung DR, et al. Risk factors and pathogenic significance of severe sepsis and septic shock in 2286 patients with gram-negative bacteremia. *J Infect.* 2011;62:26–33.
- Shono Y, Docampo MD, Peled JU, et al. Intestinal microbiota-related effects on graft-versus-host disease. *Int J Hematol.* 2015;101:428–437.
- Manzo VE, Bhatt AS. The human microbiome in hematopoiesis and hematologic disorders. *Blood.* 2015;126:311–318.
- Beelen DW, Elmaagacli A, Muller KD, et al. Influence of intestinal bacterial decontamination using metronidazole and ciprofloxacin or ciprofloxacin alone on the development of acute graft-versus-host disease after marrow transplantation in patients with hematologic malignancies: final results and long-term follow-up of an open-label prospective randomized trial. *Blood.* 1999;93: 3267–3275.
- Docampo MD, Auletta JJ, Jenq RR. Emerging Influence of the intestinal microbiota during allogeneic hematopoietic cell transplantation: control the gut and the body will follow. *Biol Blood Marrow Transplant.* 2015;21:1360–1366.